

Reduced-Order Observer-Based Electromagnetic Stress Regulation via Flux Pinning and Electrostatic Modulation

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Abstract—This paper presents a reduced-order framework for electromagnetic stress regulation using persistent flux pinning in type-II superconductors and dynamically modulated electrostatic fields. In the proposed approach, magnetic flux pinning within type-II superconducting structures provides a stable, flux-supported electromagnetic field topology that functions as a passive electromagnetic stress scaffold, while segmented electrostatic actuation provides rapid localized modulation of the Maxwell stress distribution without requiring large-scale alteration of the baseline magnetic field topology. The system is formulated directly in terms of electromagnetic stress energy regulation using the Maxwell stress tensor, thereby recasting field topology regulation as a nonlinear multiphysics control problem rather than solely a current- or voltage-regulation problem.

A reduced-order representation is developed incorporating pinned flux dynamics, electrostatic field shaping, thermal coupling, and mechanical stress interactions. Proper Orthogonal Decomposition (POD) is employed to retain the dominant electromagnetic stress modes while reducing the computational complexity associated with fully distributed multiphysics field formulations. This formulation enables computationally tractable observer-based regulation of the dominant electromagnetic stress modes while preserving the dominant dynamics.

Building upon this reduced-order formulation, a baseline Linear Quadratic Gaussian (LQG) compensator, comprising Linear Quadratic Regulator (LQR) state-feedback control and Kalman state estimation, is synthesized for observer-based regulation of the dominant electromagnetic stress coordinates. The framework is subsequently extended using Sliding Mode Control (SMC) to address nonlinear coupling, thermal drift, flux creep, and modeling uncertainty. Comparative simulations evaluate the regulation, estimation, and robustness characteristics of the LQG and SMC architectures under representative disturbance and uncertainty conditions.

The proposed framework demonstrates the feasibility of reduced-order, observer-based regulation of superconducting electromagnetic stress fields and establishes a foundation for future adaptive electromagnetic field shaping and superconducting control architectures.

Index Terms—flux pinning, electromagnetic stress regulation, Maxwell stress tensor, proper orthogonal decomposition, reduced-order modeling, linear quadratic Gaussian control, sliding mode control, type-II superconductors, vortex physics

I. INTRODUCTION

ELECTROMAGNETIC energy conversion and field-control systems based on permanent magnet architectures continue to play a dominant role in modern electric machines, high-density actuators, and electromechanical control platforms due to their high force density and efficient magnetic

flux generation. However, emerging high-energy electromagnetic systems increasingly require improved field stability, controllability, and electromagnetic stress-regulation capability under nonlinear multiphysics operating conditions. This paper investigates a control-oriented electromagnetic stress architecture combining pinned magnetic flux topologies with dynamically modulated electrostatic fields as a framework for adaptive electromagnetic stress and field topology regulation.

A. Motivation and Relation to Prior Works

Electromagnetic force generation and regulation play a central role in numerous engineering systems, including electric machines, actuators, magnetic bearings, precision positioning systems, magnetic levitation platforms, energy-conversion devices, and aerospace electromechanical subsystems. In many of these applications, the ability to produce strong, stable, and controllable electromagnetic fields directly determines system performance, efficiency, and reliability [1], [2]. Permanent magnets, for example, are among the most widely utilized technologies for generating persistent magnetic fields. Their ability to provide substantial magnetic flux without continuous power consumption has enabled widespread adoption in electric motors, generators, sensors, magnetic bearings, and precision electromechanical systems. However, their use is accompanied by several persistent challenges, including thermal demagnetization, magnetic creep and hysteresis effects, eddy current losses, nonlinear material behavior, and increasing dependence on rare-earth material supply chains. These limitations become increasingly significant in high-field, high-temperature, high-frequency, and long-duration operating environments in which field stability, controllability, and electromagnetic stress management are critical. These challenges motivate investigation into alternative electromagnetic architectures capable of improved stability and controllability under demanding operating conditions.

Type-II superconductors provide an alternative mechanism for generating persistent magnetic fields through magnetic flux quantization and flux pinning phenomena [3]–[6]. When magnetic vortices become immobilized at material defects, substantial magnetic fields may be maintained with minimal dissipation, enabling long-lived electromagnetic field topologies and potentially improved field stability [3]–[6]. Recent investigations have additionally demonstrated that electrostatic fields can influence superconducting transport properties and

critical-current behavior, motivating the possibility of combined flux pinning and electrostatic field-control architectures [7]. Electromagnetic force regulation has been extensively investigated in active magnetic bearing systems, magnetic levitation platforms, and electromagnetic actuators [1], [2], [8]. In these applications, feedback control is employed to regulate electromagnetic forces through continuous adjustment of coil currents, magnetic fields, or actuator inputs. Such systems have demonstrated high precision and dynamic performance across a broad range of engineering applications. The present work differs in that the primary electromagnetic energy storage mechanism is provided by a persistent pinned flux topology within a type-II superconductor. Rather than continuously generating the underlying magnetic field through energized coils, the proposed architecture seeks to exploit flux pinning as a passive electromagnetic stress scaffold while electrostatic actuation provides localized stress regulation. Consequently, the functions of electromagnetic energy storage and active control become partially decoupled, potentially reducing continuous power requirements and thermal loading while preserving dynamic tunability.

B. Methods and Contributions

A reduced-order formulation is developed incorporating pinned flux dynamics, electrostatic field shaping, thermal coupling, and mechanical stress interactions. Proper Orthogonal Decomposition (POD) is employed to identify and retain the dominant electromagnetic stress modes [9]–[11], yielding a reduced-order representation that substantially decreases the computational complexity associated with distributed multi-physics field formulations [12]. A baseline Linear Quadratic Gaussian (LQG) compensator, comprising Linear Quadratic Regulator (LQR) state-feedback control and observer-based state estimation [13], [14], is synthesized for regulation of the dominant electromagnetic stress coordinates. The framework is subsequently extended using Sliding Mode Control (SMC) to address nonlinear coupling, thermal drift, flux creep effects, and modeling uncertainty [15]–[17]. Comparative simulation studies evaluate the regulation, estimation, and robustness characteristics of the LQG and SMC architectures under representative disturbance and uncertainty conditions, consistent with established simulation-based validation practices for reduced-order and robust control systems [15]–[17], [21], [22]. The resulting framework establishes a foundation for simulation-driven investigation of adaptive electromagnetic stress regulation, superconducting field stabilization, and active electromagnetic field shaping.

II. PHYSICAL MODELING FRAMEWORK

A type-II superconducting domain containing a pinned Abrikosov vortex ensemble is subjected to an external magnetic bias field $\mathbf{B}_p$. Segmented electrostatic gate electrodes provide actuation for modulation of the resulting electromagnetic stress distribution. Hall-effect sensing is shown as one possible measurement mechanism for future closed-loop

implementations. Vortex locations are schematic and are not intended to represent actual vortex density or lattice structure.

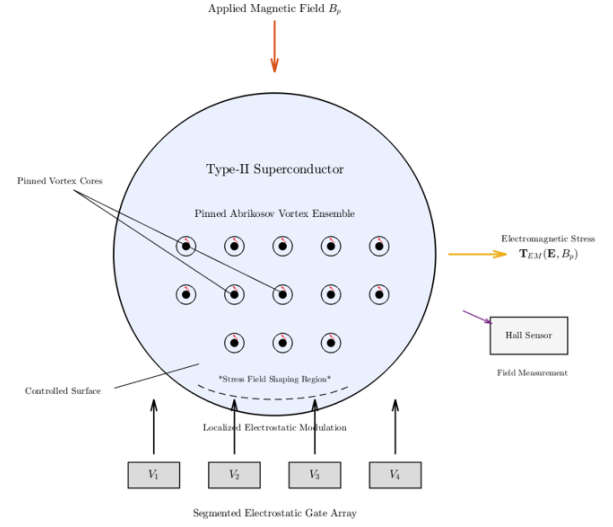


Fig. 1. Conceptual architecture of the proposed pinned flux electromagnetic stress regulation system

A. Maxwell Stress Tensor

The electromagnetic stress state is represented through the Maxwell stress tensor [1], given by

$$\mathbf{T}_{EM} = \epsilon \left(\mathbf{E}\mathbf{E} - \frac{1}{2}E^2\mathbf{I} \right) + \frac{1}{\mu} \left(\mathbf{B}\mathbf{B} - \frac{1}{2}B^2\mathbf{I} \right) \quad (1)$$

where ϵ and μ denote the permittivity and permeability of the medium, respectively, and $\mathbf{I}$ is the identity tensor. The electromagnetic stress state of the system is formulated using the Maxwell stress tensor, which provides a continuum representation of the local electromagnetic force density and momentum transfer associated with coupled electric and magnetic fields [1]. In the present framework, the Maxwell stress tensor is adopted as the primary quantity that governs the regulation of electromagnetic stress, enabling the field topology to be represented directly in terms of distributed electromagnetic pressure and shear interactions rather than solely through current or voltage variables.

The electric field contribution represents electrostatic pressure and field-induced normal stress, while the magnetic field contribution captures magnetic tension and transverse magnetic pressure associated with the pinned flux topology. The resulting tensor field therefore provides a compact description of the coupled electromagnetic stress distribution governing the dominant reduced-order dynamics considered throughout this work.

B. Force from Maxwell Stress Tensor

$$\mathbf{F} = \oint_S \mathbf{T}_{EM} \cdot \hat{\mathbf{n}} dS \quad (2)$$

The net electromagnetic force acting on the controlled region is obtained through surface integration of the Maxwell stress tensor over the boundary surface S [1], [3], where $\hat{\mathbf{n}}$ denotes the outward unit normal vector. This formulation enables distributed electromagnetic stress interactions to be represented as an equivalent resultant force acting on the reduced-order modal coordinates governing the pinned flux dynamics. Within the proposed architecture, the surface integral representation provides a natural bridge between the distributed electromagnetic field formulation and the reduced-order control-oriented state-space representation developed in the subsequent sections. In particular, perturbations in the electric field, pinned magnetic flux topology, and electrostatic actuation modify the local Maxwell stress distribution, thereby altering the net electromagnetic forcing applied to the retained reduced-order modes.

C. Reduced-Order Modal Representation

The full electromagnetic stress distribution associated with the proposed pinned flux architecture is inherently spatially distributed and governed by coupled electromagnetic, thermal, and mechanical interactions. Direct solution of the fully distributed multiphysics field equations is computationally expensive and generally unsuitable for real-time control-oriented analysis. Consequently, a reduced-order representation is adopted to capture the dominant electromagnetic stress dynamics while retaining computational tractability.

A snapshot-based Proper Orthogonal Decomposition (POD) framework is employed to extract the dominant electromagnetic stress modes from the distributed Maxwell stress field [8]:

Let us denote by $\tilde{\mathbf{X}}$ the centered snapshot matrix constructed from sampled electromagnetic stress distributions over the controlled surface [8]. Singular Value Decomposition (SVD) is then applied, yielding

$$\tilde{\mathbf{X}} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T \quad (3)$$

where $\mathbf{U}$ contains the orthogonal spatial POD modes, $\mathbf{\Sigma}$ contains the singular values, and $\mathbf{V}$ contains the temporal modal coefficients. These singular values quantify the relative energetic contribution of each mode to the overall electromagnetic stress response. The normalized modal energy contribution associated with the i^{th} mode is defined as

$$\eta_i = \frac{\sigma_i^2}{\sum_j \sigma_j^2} \quad (4)$$

where σ_i denotes the i^{th} singular value.

Modal truncation is subsequently performed by retaining only the smallest subset of modes satisfying a prescribed cumulative stress energy threshold,

$$\sum_{i=1}^{N_r} \eta_i \geq \eta_{req} \quad (5)$$

where N_r denotes the number of retained modes and η_{req} is the required cumulative energy fraction. In practice, values of η_{req} between 90% and 99% are commonly employed to retain the dominant energetic content of the system while reducing computational complexity [8], [12]. In the present formulation, N_r is selected as the minimum number of modes satisfying (5), ensuring that the dominant electromagnetic stress dynamics are retained while higher-order modes contributing negligibly to the overall stress variance are discarded [8], [12].

The retained modes correspond to the dominant electromagnetic stress dynamics relevant to closed-loop field regulation. These include large-scale pinned flux displacement modes, dipole and quadrupole stress asymmetries, slow thermal drift modes, and low-order field topology perturbations. Higher-order spatial harmonics, localized fringe-field effects, microscopic vortex fluctuations, and high-frequency electromagnetic wave phenomena are truncated under the assumption that their contribution to the controlled stress energy dynamics is comparatively small within our intended operating regime.

The resulting reduced-order modal coordinates form the basis for the state-space modeling, observer design, and nonlinear electromagnetic stress regulation in the subsequent sections.

D. Pinned Flux Displacement

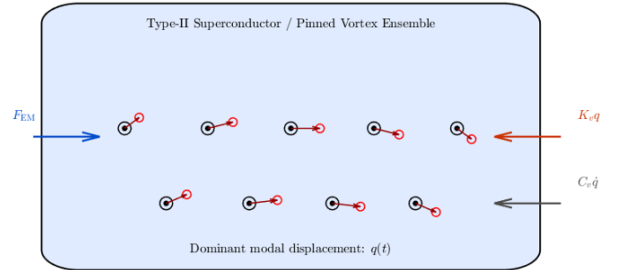


Fig. 2. Reduced-order representation of the pinned flux displacement mode

We model perturbations of the pinned flux topology using a reduced-order modal representation. The detailed POD-based modal decomposition is introduced in Section VI-C. For the present development, let q denote a generic retained modal coordinate:

$$M_v \ddot{q} + C_v \dot{q} + K_v q = \mathbf{F}_{EM} + d \quad (6)$$

where q denotes a pinned flux modal coordinate, while M_v , C_v , and K_v represent the modal inertia, damping, and restoring stiffness, respectively, associated with the vortex dynamics. The disturbance term d represents unmodeled multiphysics effects, including thermal drift, flux creep, localized vortex fluctuations, and external perturbations. The controllable electromagnetic forcing term $\mathbf{F}_{EM}$ representing electromagnetic loading generated by the coupled magnetic and electrostatic fields is modeled as

$$\mathbf{F}_{EM} = \alpha_B \Phi^2 + \alpha_E V_e^2 \quad (7)$$

where Φ denotes the pinned magnetic flux state, V_e is the effective electrostatic actuator voltage, and α_B and α_E are

electromagnetic coupling coefficients representing magnetic and electrostatic stress contributions, respectively. The coefficient α_E therefore represents an effective electrostatic coupling parameter associated with experimentally observed electric field-induced modifications of superconducting vortex behavior [7].

This reduced-order representation is adopted by decomposing the spatially distributed Maxwell stress response into a manageable set of dominant coupled electromagnetic, thermal, and mechanical modes relevant to field-stress regulation. This approach is intended to reduce the computational complexity associated with fully distributed multiphysics field formulations while preserving the dominant observable dynamics governing electromagnetic stress evolution. Since the primary control objective is regulation of electromagnetic stress rather than reconstruction of the complete electromagnetic field distribution, the modal basis is constructed from snapshots of the normal Maxwell pressure and stress tensor components evaluated over the controlled surface.

Proper Orthogonal Decomposition (POD) is then applied to the centered snapshot matrix, and the retained modal coordinates are selected according to their cumulative stress energy contribution. Modes associated with baseline pinned flux topology, dipole asymmetry, quadrupole distortion, axial gap variation, and slow thermal drift are retained, while high-order spatial harmonics, microscopic vortex motion, local fringe-field effects, and high-frequency electromagnetic wave dynamics are truncated.

III. PHYSICAL MECHANICS OF THE PINNED FLUX STATE

A. Ginzburg-Landau Framework and Scale Separation

It should be emphasized that this work does not seek a first-principles microscopic solution of the Ginzburg-Landau equations. Rather, the proposed framework adopts a control-oriented reduced-order representation suitable for electromagnetic stress regulation. Microscopic superconducting effects are incorporated implicitly through reduced-order coupling coefficients governing the macroscopic electromagnetic stress dynamics. Consequently, the governing equations should be interpreted as macroscopic reduced-order approximations of the collective dynamics of pinned vortex ensembles rather than direct microscopic solutions of the Ginzburg-Landau equations.

B. Abrikosov Vortex Quantization

When the applied magnetic field lies between the lower and upper critical fields, that is,

$$\mathbf{H}_{c1} < \mathbf{H} < \mathbf{H}_{c2},$$

magnetic flux penetrates a type-II superconductor in the form of quantized vortex lines rather than as a continuous field distribution [1]. The characteristic superconducting length scales are the London penetration depth λ and the coherence

length ξ [9]. Their ratio defines the Ginzburg-Landau parameter:

$$\kappa = \frac{\lambda}{\xi}.$$

A material exhibits Type-II superconductivity when

$$\kappa > \frac{1}{\sqrt{2}},$$

for which magnetic flux penetrates the material in the form of quantized Abrikosov vortices [1], [3]. Each vortex consists of a normal-conducting core of characteristic radius on the order of ξ , surrounded by circulating supercurrents that decay over the London penetration depth [9].

The total magnetic flux associated with each vortex is quantized according to

$$\Phi = n \Phi_0,$$

with

$$\Phi_0 = \frac{h}{2e},$$

where h denotes Planck's constant, and e is the elementary charge [1], [9].

Consequently, the magnetic flux configuration of a type-II superconductor may be interpreted as a discrete collection of flux quanta whose spatial arrangement is governed by vortex-vortex interactions, pinning sites, and externally applied fields [1], [3]. In the present work, the pinned flux state variable Φ_p represents a macroscopic aggregate description of these quantized vortex structures. The resulting flux topology acts as a persistent passive electromagnetic stress scaffold upon which electrostatic modulation and closed-loop stress regulation are imposed. Quantized flux penetration and vortex-lattice formation therefore provide the physical basis for treating the baseline magnetic topology as a stable reduced-order state [1], [3].

C. Microscopic Origin of the Damped Elastic Mode

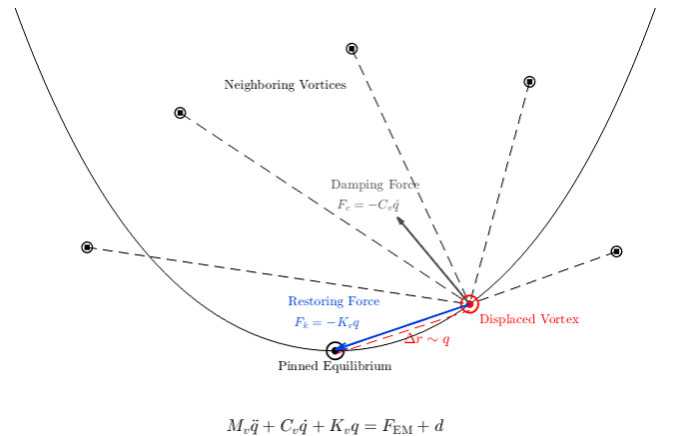


Fig. 3. Microscopic origin of the damped elastic vortex mode

A vortex displaced from its pinned equilibrium position experiences restoring forces arising from the pinning potential

and vortex-vortex interactions. Dissipative flux motion introduces damping, producing the effective reduced-order modal dynamics used in Equation (6). Although vortex dynamics originate from microscopic superconducting physics, the collective motion of pinned vortices can often be approximated as a damped elastic system [4], [24]. Small perturbations of a vortex from its equilibrium pinning site generate restoring forces associated with the local pinning potential, while vortex motion through the superconducting medium produces viscous dissipation [4], [24]. The reduced-order pinned flux dynamics introduced in (6) may be physically interpreted through established models of vortex motion in type-II superconductors. Namely, the coefficients M_v , C_v , and K_v correspond to the vortex inertia, Bardeen-Stephen viscous drag, and the Labusch pinning stiffness governing the restoring force of the pinning potential wells, respectively [4], [24].

IV. BASELINE CONTROL PROBLEM FORMULATION

A. State-Space Representation

Let the reduced state vector be

$$\mathbf{x}(t) = \begin{bmatrix} q(t) \\ \dot{q}(t) \\ \Phi_p(t) \\ T(t) \\ V_e(t) \end{bmatrix} \quad (8)$$

where $q(t)$ denotes the retained modal coordinate used for control synthesis, $\dot{q}(t)$ denotes the modal velocity, Φ_p is the pinned flux state, T is the thermal state, and V_e is the equivalent electrostatic actuation state.

The nonlinear reduced-order dynamics are written as

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, u) + \mathbf{w} \quad (9)$$

with

$$\mathbf{f}(\mathbf{x}, u) = \begin{bmatrix} \dot{q} \\ \frac{1}{M_v} (\alpha_B \Phi_p^2 + \alpha_E V_e^2 - C_v \dot{q} - K_v q + d) \\ -\gamma_\Phi (\Phi_p - \Phi_{p,0}) \\ \frac{1}{C_{th}} (k_E V_e^2 + k_\Phi \Phi_p^2 - h(T - T_0)) \\ \frac{1}{\tau_e} (-V_e + u) \end{bmatrix} \quad (10)$$

and $\mathbf{w}$ is the process noise. The final component of $\mathbf{f}(\mathbf{x}, u)$ represents first-order electrostatic actuation dynamics, where u is the commanded equivalent electrostatic input and τ_e is the corresponding actuation time constant.

The measurement equation is

$$\mathbf{y} = \mathbf{g}(\mathbf{x}) + \mathbf{v} \quad (11)$$

with

$$\mathbf{g}(\mathbf{x}) = \begin{bmatrix} q \\ \Phi_p \\ T \\ V_e \end{bmatrix}, \quad (12)$$

and $\mathbf{v}$ is measurement noise.

Although practical implementations may employ arrays of independently addressable segmented electrodes, with individual voltages represented by

$$\mathbf{V}_{seg} = [V_{e1} \ V_{e2} \ \dots]^T,$$

the present work represents their collective action by a single equivalent electrostatic control input V_e . This reduced-order approximation captures the aggregate influence of electrostatic actuation on the retained electromagnetic stress modes while avoiding explicit modeling of individual electrode dynamics.

For the LQG design, the nonlinear dynamics are linearized around the operating point

$$x_0 = \begin{bmatrix} q_0 \\ 0 \\ \Phi_{p,0} \\ T_0 \\ V_{e,0} \end{bmatrix}, \quad u_0 = V_{e,0}. \quad (13)$$

Using perturbation variables

$$\delta \mathbf{x} = \mathbf{x} - x_0, \text{ and} \\ \delta u = u - u_0$$

we obtain

$$\delta \dot{\mathbf{x}} = \mathbf{A} \delta \mathbf{x} + \mathbf{B} \delta u + \mathbf{w}, \quad (14)$$

$$\delta \mathbf{y} = \mathbf{C} \delta \mathbf{x} + \mathbf{v}. \quad (15)$$

The linearized system matrix is

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ -\frac{K_v}{M_v} & -\frac{C_v}{M_v} & \frac{2\alpha_B \Phi_{p,0}}{M_v} & 0 & \frac{2\alpha_E V_{e,0}}{M_v} \\ 0 & 0 & -\gamma_\Phi & 0 & 0 \\ 0 & 0 & \frac{2k_\Phi \Phi_{p,0}}{C_{th}} & -\frac{h}{C_{th}} & \frac{2k_E V_{e,0}}{C_{th}} \\ 0 & 0 & 0 & 0 & -\tau_e^{-1} \end{bmatrix} \quad (16)$$

The off-diagonal coupling terms explicitly capture the interaction between vortex-state dynamics, thermal behavior, and electrostatic actuation. In particular, the terms proportional to α_B , α_E , k_Φ , and k_E describe electromagnetic and thermo-electromagnetic coupling introduced by the pinned flux and electrostatic subsystems.

The input matrix is

$$\mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ \tau_e^{-1} \end{bmatrix} \quad (17)$$

The measured outputs are q , Φ_p , T , and V_e , so

$$\mathbf{C} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{D} = \mathbf{0}. \quad (18)$$

The LQR control law is

$$\delta \mathbf{u} = -\mathbf{K}_{\text{LQR}} \delta \hat{\mathbf{x}}, \quad (19)$$

and the Kalman state estimator is

$$\dot{\delta \hat{\mathbf{x}}} = \mathbf{A} \delta \hat{\mathbf{x}} + \mathbf{B} \delta \mathbf{u} + \mathbf{L} (\delta \mathbf{y} - \mathbf{C} \delta \hat{\mathbf{x}}) \quad (20)$$

Finally, the control objective is to regulate the electromagnetic stress state of the pinned flux system toward a desired equilibrium configuration by minimizing the deviation between the generated Maxwell stress tensor and a reference stress distribution:

$$\min_{\mathbf{u}} \|\mathbf{T}_{\text{EM}}(\mathbf{E}(\mathbf{u}), \mathbf{B}_p) - \mathbf{T}_{\text{ref}}\|^2 \quad (21)$$

B. Controllability and Observability Analysis

The controllability and observability properties of the linearized reduced-order model were evaluated through the controllability matrix

$$\mathbf{P}_c = [\mathbf{B} \quad \mathbf{A}\mathbf{B} \quad \mathbf{A}^2\mathbf{B} \quad \dots \quad \mathbf{A}^{n-1}\mathbf{B}]$$

and the observability matrix

$$\mathbf{P}_o = [\mathbf{C} \quad \mathbf{C}\mathbf{A} \quad \mathbf{C}\mathbf{A}^2 \quad \dots \quad \mathbf{C}\mathbf{A}^{n-1}]^T.$$

Evaluation of the corresponding matrix ranks indicates that the complete linearized model is not fully controllable; that is, $\text{rank}(\mathbf{P}_c) = 4$, indicating the existence of a one-dimensional uncontrollable subspace. The eigenvalue associated with this subspace corresponds to the autonomous flux relaxation dynamics and lies in the open left-half plane, so although the system is not fully controllable, it remains stabilizable.

Since the pair $(\mathbf{A}, \mathbf{B})$ is stabilizable and the pair $(\mathbf{A}, \mathbf{C})$ is observable ($\text{rank}(\mathbf{P}_o) = 5$) under the assumed measurement set, compensator-based output feedback may be applied to the regulated, reduced-order subsystem, with the pinned flux state entering the controller as an estimated scheduling variable. As a practical consequence, direct measurement of the full reduced-order state vector is not required, and implementation may be realized using experimentally accessible outputs such as displacement, stress, magnetic field, or capacitive sensing measurements.

V. EXTENSION TO SLIDING-MODE CONTROL

The original Maxwell stress minimization objective remains the global regulation objective. In the SMC formulation, this objective is projected onto the retained reduced-order stress

coordinates, and the controller is designed to drive the corresponding sliding variables to zero under bounded uncertainty.

We define

$$e(t) = q(t) - q_{\text{ref}}(t) \quad (22)$$

where $q_{\text{ref}}(t)$ is the desired dominant Maxwell stress/asymmetry modal coordinate value. For the present investigation, the desired reduced-order stress coordinate is selected as $q_{\text{ref}} = 0$, which corresponds to regulation about the nominal pinned flux equilibrium topology.

Consistent with the stabilizability and observability properties established in Section IV-B, the reduced-order state vector may be reconstructed from the available output measurements through observer-based estimation [13], allowing implementation using only measurable outputs such as displacement, stress, magnetic field, or capacitive sensing signals. Alternative robust observer architectures, including sliding-mode observers for nonlinear uncertain systems, have been widely investigated in the literature [18], [20].

We then define the sliding surface

$$\mathbf{s}(t) = \dot{\mathbf{e}}(t) + \eta \mathbf{e}(t) \quad (23)$$

with $\eta > 0$.

This first-order sliding surface was selected according to the conventional sliding-mode formulation [4]. The objective then becomes

$$\mathbf{s}(t) \rightarrow \mathbf{0}$$

which implies

$$\mathbf{e}(t) \rightarrow \mathbf{0}.$$

Thus, rather than directly minimizing the full Maxwell stress distribution at every instant, the controller regulates the dominant reduced-order stress mode, since the retained modes capture the majority of the electromagnetic stress variance present in the full field distribution [8], [12].

The SMC control law is chosen as

$$u_{\text{smc}} = -\mathbf{K}_s \text{sat}\left(\frac{\mathbf{s}}{\varphi}\right) \quad (24)$$

To reduce high-frequency switching and mitigate chattering-induced thermal and electromagnetic stress oscillations, the discontinuous sign function typical of a classical SMC formulation is replaced by a continuous saturation function introducing a thin boundary layer around the sliding manifold [4]:

$$\text{sat}\left(\frac{\mathbf{s}}{\varphi}\right) = \begin{cases} 1, & \mathbf{s} > \varphi \\ \frac{\mathbf{s}}{\varphi}, & |\mathbf{s}| \leq \varphi \\ -1, & \mathbf{s} < -\varphi \end{cases} \quad (25)$$

where $\mathbf{s}$ is the sliding surface and φ is the boundary layer thickness. This simple modification will ensure smoother transitions in the vicinity of $\mathbf{s} = 0$.

Physically, the boundary layer may be interpreted as a finite tolerance region around the desired electromagnetic stress equilibrium in which small residual tracking errors are permitted in exchange for improved robustness and reduced actuator stress. Rather than continuously forcing the system to switch across the ideal sliding manifold, the controller allows bounded motion within a narrow neighborhood of the manifold, thereby preventing unnecessary electrostatic actuation and reducing the risk of high-frequency excitation of unmodeled electromagnetic, thermal, or vortex-dynamic phenomena.

The equivalent control is chosen to cancel the known nominal dynamics, while the switching term rejects uncertainty:

$$V_e^2 = \frac{1}{\alpha_E} \left[M \left(\ddot{q}_{\text{ref}} - \eta \dot{e} - \mathbf{K}_s \text{sat} \left(\frac{s}{\varphi} \right) \right) + C_v \dot{q} + K_v q - \alpha_B \Phi_p^2 \right]. \quad (26)$$

Substituting (26) into the reduced-order dynamics yields

$$\dot{s} = -\mathbf{K}_s \text{sat} \left(\frac{s}{\varphi} \right) + \Delta(\mathbf{x}, t), \quad (27)$$

where $\Delta(x, t)$ denotes the lumped uncertainty arising from unmodeled electromagnetic coupling, thermal drift, flux-creep effects, and parametric mismatch.

The desired electrostatic command is subsequently limited according to the actuator constraints:

$$|V_e| \leq V_{max} \quad (28)$$

$$T < T_c \quad (29)$$

$$H < H_{c2} \quad (30)$$

Equation (28) represents the maximum available electrostatic actuation voltage, while (29) and (30) ensure that the superconducting material remains below the critical temperature and the magnitude of the applied magnetic field remains lower than the upper critical magnetic field strength, respectively, thereby avoiding quench conditions and loss of flux pinning.

It must be noted so that the electrostatic actuator voltage V_e not be confused with the Lyapunov function V_s introduced below: The quantity V_e represents a physical control input applied through the segmented electrode array, whereas V_s is a mathematical measure of closed-loop stability.

To establish stability of the sliding dynamics, let us consider the standard quadratic Lyapunov candidate function [4]:

$$V_s = \frac{1}{2} \mathbf{s}^2 \quad (31)$$

Differentiating yields

$$\dot{V}_s = \mathbf{s} \dot{\mathbf{s}}. \quad (32)$$

Assume that lumped uncertainty entering the reduced-order stress dynamics be bounded by

$$|\Delta(\mathbf{x}, t)| \leq \Delta_{\max}. \quad (33)$$

and choose gain $\mathbf{K}_s$ such that $\mathbf{K}_s > \Delta_{\max}$.

Multiplying by $\mathbf{s}$ yields

$$\mathbf{s} \dot{\mathbf{s}} = -\mathbf{K}_s \mathbf{s} \text{sgn}(\mathbf{s}) + \mathbf{s} \Delta(\mathbf{x}, t). \quad (34)$$

Then, using the identity

$$\mathbf{s} \text{sgn}(\mathbf{s}) = |\mathbf{s}| \quad (35)$$

and the uncertainty bound

$$|\mathbf{s} \Delta(\mathbf{x}, t)| \leq |\mathbf{s}| \Delta_{\max}, \quad (36)$$

it follows that

$$\mathbf{s} \dot{\mathbf{s}} \leq -(\mathbf{K}_s - \Delta_{\max}) |\mathbf{s}| < 0, \quad \forall \mathbf{s} \neq 0 \quad (37)$$

meaning that $\dot{V}_s < 0$.

Therefore, V_s decreases monotonically along the closed-loop trajectories, implying convergence of the reduced-order stress dynamics toward the sliding manifold.

Because the boundary layer approximation replaces the ideal discontinuous switching law, convergence occurs to a bounded neighborhood of the manifold whose size is determined by the parameter φ . Consequently, the tracking error remains uniformly bounded and can be made arbitrarily small through appropriate selection of the boundary-layer thickness.

From a physical perspective, this result implies that perturbations of the dominant electromagnetic stress mode are actively suppressed despite the presence of thermal drift, flux creep effects, parametric uncertainty, and external disturbances. The controller therefore acts to restore the pinned flux configuration toward its nominal equilibrium topology while maintaining admissible actuator and material operating conditions.

The resulting SMC formulation preserves the original objective of electromagnetic stress regulation while providing improved robustness relative to the baseline LQG architecture. In particular, the switching action compensates for uncertainty in the reduced-order electromagnetic stress model and reduces sensitivity to modeling errors associated with vortex dynamics, thermal coupling, and nonlinear electromagnetic interactions. The effectiveness of the proposed reduced-order SMC architecture is investigated in the following section using representative electromagnetic stress field simulations and Monte Carlo uncertainty analysis.

VI. SIMULATION GEOMETRY AND METHODOLOGY

A. Normalized stress field Geometry

The simulated controlled surface is represented in $(\mathbf{r}, \theta)$, where $0 \leq \mathbf{r} \leq 1$ and $0 \leq \theta < 2\pi$.

The normal Maxwell pressure $\mathbf{P}(\mathbf{r}, \theta, t)$ is modeled as a normalized baseline pinned flux pressure with superimposed low-order spatial perturbations representing dipole asymmetry, quadrupole distortion, radial/gap-like variation, thermal drift, and measurement noise. These amplitudes are selected as representative normalized perturbation levels for preliminary reduced-order simulation and may be replaced by experimentally identified or high-fidelity FEM-derived modal amplitudes in future work.

B. Snapshot Generation

The electromagnetic stress field is represented by a sequence of time-dependent spatial snapshots generated over the simulation interval. Each snapshot corresponds to a discretized realization of the normal Maxwell stress distribution $\mathbf{P}(\mathbf{r}, \theta, t)$ across the controlled surface. The resulting collection of snapshots forms the matrix

$$\mathbf{X} = [\mathbf{x}_1 \quad \mathbf{x}_2 \quad \cdots \quad \mathbf{x}_N]$$

where each column $\mathbf{x}_k$ contains the spatial stress field at time t_k . The simulated stress distribution consists of a nominal pinned flux equilibrium pressure together with superimposed perturbations representing dipole asymmetry, quadrupole distortion, radial deformation, thermal drift, and measurement noise. These perturbations are intended to emulate physically plausible deviations from an ideal pinned flux configuration while providing sufficient spatial and temporal variation for reduced-order analysis.

Prior to decomposition, the snapshot matrix is centered according to

$$\tilde{\mathbf{X}} = \mathbf{X} - \bar{\mathbf{X}},$$

where $\bar{\mathbf{X}}$ denotes the mean stress field. Centering removes the static baseline component and ensures that the retained POD modes represent dominant electromagnetic stress perturbations about the nominal pinned flux equilibrium rather than the equilibrium itself. Consequently, the resulting reduced-order coordinates describe the dynamically significant stress variations that are most relevant for feedback control and disturbance rejection.

The present investigation is intended to establish a reduced-order control framework rather than a calibrated device-specific model. For this reason, the parameters selected are normalized to represent physically plausible relative perturbation magnitudes while preserving the expected modal hierarchy of electromagnetic stress asymmetries. The resulting simulations should therefore be interpreted as proof-of-concept demonstrations of the proposed control architecture rather than quantitative predictions for a specific superconducting device.

C. POD-Based Reduction

The retained POD modes found from (3) constitute an empirical reduced-order basis describing the dominant electromagnetic stress perturbations present within the snapshot ensemble. Projection of the full-order stress field onto this basis yields the reduced-order coordinates

$$\mathbf{q}_r(t) = \mathbf{U}_r^T \tilde{\mathbf{X}}(t), \quad (38)$$

which provide a compact representation of the dominant electromagnetic stress dynamics. Thus, rather than describing the full distributed stress field directly, the reduced-order representation captures the majority of the stress-field energy using a small number of modal coordinates. Although the first three POD modes are retained to characterize the dominant electromagnetic stress field, the present work restricts feedback control to the leading modal coordinate. Since this mode captures 85.2% of the electromagnetic stress energy in the representative snapshot ensemble, single-mode regulation provides an appropriate first demonstration of the proposed reduced-order control framework. Extension to simultaneous regulation of the full retained modal vector is reserved for future investigation.

D. Monte Carlo Uncertainty Sweep

To assess robustness and obtain a conservative estimate of model uncertainty, a Monte Carlo parameter sweep is performed [21], [22]. The baseline pinned flux pressure, modal amplitudes, thermal drift coefficient, and measurement-noise level are varied independently within prescribed uncertainty intervals representing plausible deviations from nominal operating conditions. For each realization, a new snapshot ensemble is generated and projected onto the nominal retained POD basis. The resulting reduced-order trajectories are compared against the nominal reduced-order dynamics to quantify the mismatch induced by parameter uncertainty and stochastic disturbances. In particular, the modal acceleration discrepancy

$$\Delta(t) = \ddot{q}_{mc}(t) - \ddot{q}_{nom}(t) \quad (39)$$

is evaluated throughout the simulation horizon. The maximum observed discrepancy over all realizations is used to estimate a conservative lumped uncertainty bound

$$\Delta_{max} = \max |\Delta(t)|.$$

By construction, Δ_{max} represents an empirical upper bound on the retained-mode acceleration uncertainty induced by the considered parameter variations and stochastic disturbances and therefore serves as the uncertainty estimate employed in the sliding-mode control design. The switching gain $\mathbf{K}_s$ is subsequently selected such that $\mathbf{K}_s > \Delta_{max}$, consistent with standard sliding-mode robustness conditions [4], [12], thereby ensuring that the switching term dominates bounded disturbances, parameter variation, thermal drift, and unmodeled multiphysics effects.

Supplementary Monte Carlo sensitivity studies were also performed by varying the LQR weighting matrix $\mathbf{Q}$ and

the SMC parameters η and φ . The selected nominal values correspond to regions exhibiting uniformly stable closed-loop behavior across the uncertainty ensemble, indicating that the reported results are representative rather than dependent upon a single finely tuned parameter set.

E. Simulation Parameters

The simulations presented in this work employ normalized illustrative parameter values intended to demonstrate the proposed reduced-order modeling, uncertainty quantification, and controller design methodology. The parameters are not derived from a specific superconducting device and should therefore be interpreted as dimensionless modal amplitudes rather than experimentally calibrated physical quantities.

The baseline pinned flux pressure is normalized as $P_0 = 1$. The remaining modal amplitudes are expressed as fractions of this baseline stress level: $a_{\text{sym}} = 0.20$, $q_d = 0.12$, $a_{\text{rm}} = 0.06$, and $A_n = 0.01$. These values correspond to dipole asymmetry, quadrupole distortion, axial-radial modal perturbation, and measurement-noise amplitude, respectively. This hierarchy preserves a dominant baseline pinned flux stress distribution while introducing representative low-order electromagnetic stress distortions. For the uncertainty study, $N_{\text{MC}} = 1000$ Monte Carlo realizations were performed.

TABLE I
NORMALIZED SIMULATION PARAMETERS

Parameter	Description	Value
P_0	Baseline pinned flux pressure	1.00
a_{sym}	Dipole asymmetry amplitude	0.30
q_d	Quadrupole distortion amplitude	0.12
a_{rm}	Axial-radial mode amplitude	0.06
A_n	Measurement-noise amplitude	0.01
N_{MC}	Monte Carlo realizations	1000

VII. RESULTS

The proposed reduced-order electromagnetic stress regulation framework was evaluated using a representative simulation model constructed from synthetic Maxwell stress distributions containing baseline pinned flux pressure, dipole asymmetry, quadrupole distortion, axial-radial modal perturbations, thermal drift, and stochastic measurement noise. The resulting stress fields were used to generate snapshot ensembles for Proper Orthogonal Decomposition (POD), reduced-order model construction, and uncertainty quantification.

The following results illustrate the progression from the full distributed electromagnetic stress field to the reduced-order modal representation and, subsequently, to the uncertainty estimates employed for robust sliding-mode controller design.

A. Representative Electromagnetic Stress Distribution

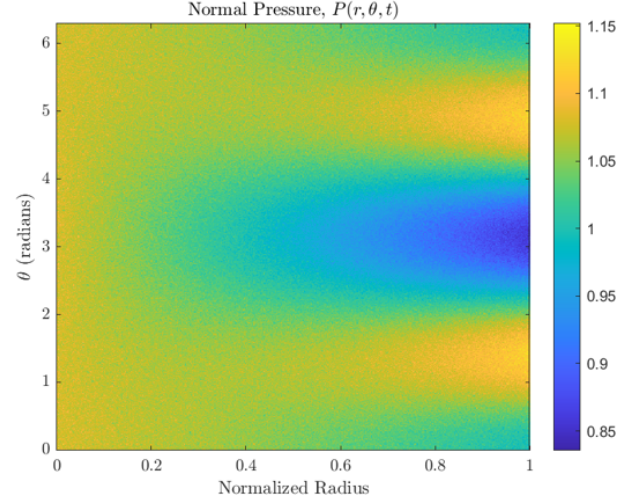


Fig. 4. Normal stress distribution over the controlled surface

A representative realization of the normal Maxwell stress distribution $\mathbf{P}(r, \theta, t)$ is shown in Figure 4. The baseline pinned flux pressure distribution is perturbed by dipole asymmetry, quadrupole distortion, axial-radial deformation, thermal drift, and low-level stochastic measurement noise. These contributions collectively produce a spatially varying electromagnetic stress field representative of the disturbances considered throughout this investigation. Several important features are immediately apparent: First, the stress field is dominated by large-scale spatial structures extending over significant portions of the controlled surface. Second, the observed variations occur primarily through smooth low-order distortions rather than highly localized fluctuations. Finally, the stochastic perturbations remain comparatively small relative to the dominant stress distribution and do not significantly alter the overall field topology.

These observations suggest that the majority of the electromagnetic stress energy content is concentrated within a relatively small number of dominant spatial modes. Consequently, a compact reduced-order representation may be expected to capture the principal stress dynamics while discarding higher-order perturbations that contribute comparatively little to the overall stress response. This hypothesis is investigated quantitatively through the POD analysis presented in the following sections.

B. Singular Value Spectrum and Modal Energy Retention

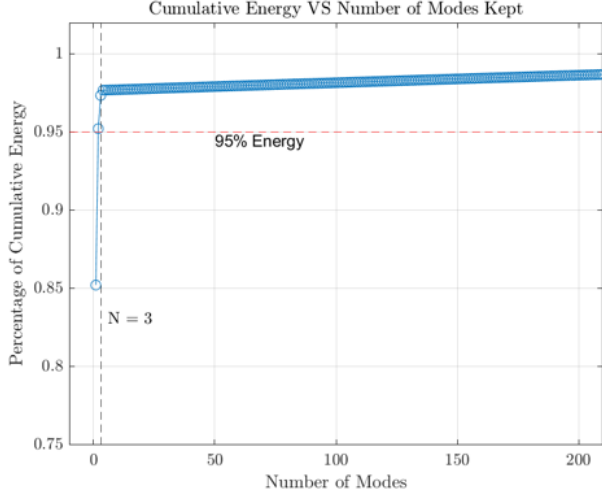


Fig. 5. Cumulative stress energy captured, as a function of number of retained POD modes

Figure 5 quantifies the cumulative stress energy retained as additional POD modes are included in the reduced-order basis. The cumulative energy rises rapidly within the first several modes and exceeds the prescribed retention threshold of 95% after only three retained modes. Thus, $N_r = 3$ is selected as the reduced-order dimension for subsequent state-space modeling and control design. The results indicate that the majority of the observable electromagnetic stress dynamics can be represented using only three dominant modal coordinates. Higher-order modes contribute comparatively little additional stress energy content and are therefore neglected in the reduced-order formulation. This substantial dimensionality reduction supports the central premise of the proposed framework: Namely, the dominant electromagnetic stress behavior can be regulated through a compact set of reduced-order coordinates, rather than the full distributed field representation.

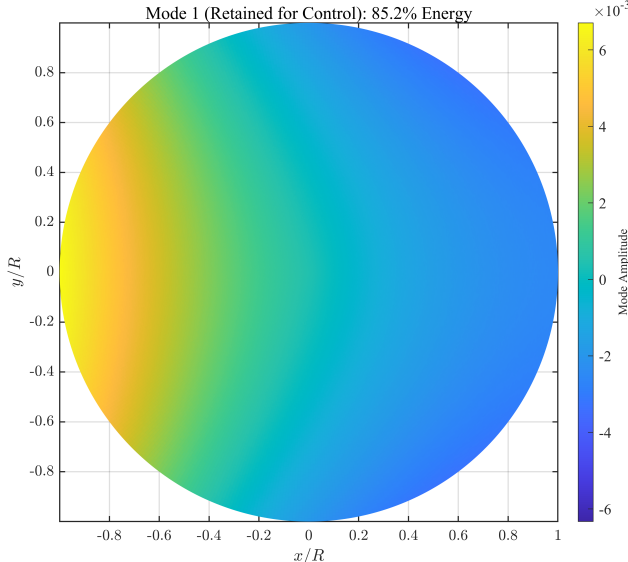


Fig. 6. The most dominant POD spatial electromagnetic stress mode

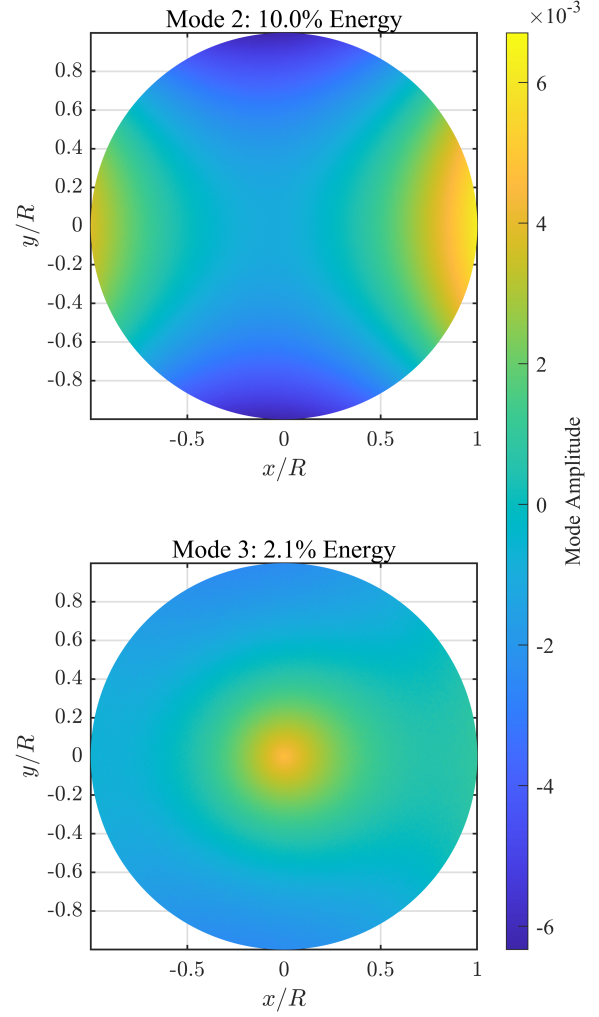


Fig. 7. The second- and third-most dominant POD spatial electromagnetic stress modes

We find energy contributions of 85.2%, 10.0%, and 2.1%, corresponding to a cumulative retained energy of 97.3%. Mode 1 captures the dominant dipole-like stress asymmetry introduced in the representative snapshot ensemble, while Modes 2 and 3 capture progressively higher-order spatial distortion patterns. This energy ordering is consistent with the Proper Orthogonal Decomposition methodology, in which the leading modes capture the dominant coherent structures of the snapshot ensemble while higher-order modes contribute progressively less energy to the reduced-order representation [23]. Consequently, the first three POD modes are retained for stress field reconstruction and modal-energy characterization. Consistent with the reduced-order formulation introduced in Section VI-C, the subsequent control synthesis focuses on the dominant POD coordinate $q_1(t)$.

C. Modal Dynamics

The retained POD modes define the reduced-order state coordinates used throughout the subsequent observer and controller development. Examination of the modal coefficient trajectories provides insight into the temporal dynamics captured by the reduced-order representation and further illustrates the

hierarchy of modal importance identified through the singular-value analysis.

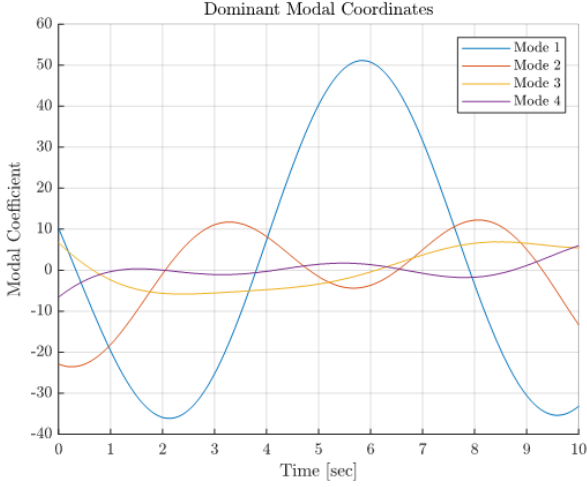


Fig. 8. Temporal evolution of the POD modal coefficients associated with the retained Maxwell stress modes

Figure 8 presents the temporal evolution of the retained modal coefficients associated with the dominant electromagnetic stress modes. This observation is consistent with the modal energy distribution reported in the previous section, where the first retained mode accounts for 85.2% of the total retained stress energy. The remaining retained modes therefore act primarily as correction terms to the dominant stress distribution, capturing secondary quadrupole-type distortions, axial-radial variations, and localized modifications of the electromagnetic stress field that are not represented by the leading mode alone.

Although several retained modes exhibit oscillatory behavior over the simulation interval, the modal trajectories remain dominated by relatively slow temporal variations associated with the imposed stress field perturbations and reduced-order operating dynamics. This behavior further supports the assumption that the dominant electromagnetic stress response can be represented through a low-dimensional set of slowly varying modal coordinates suitable for observer-based estimation and feedback regulation. From a controls perspective, the modal amplitude hierarchy has important implications for controller synthesis. Because the first retained mode contains the majority of the observable stress energy, regulation of the dominant electromagnetic stress coordinate is expected to provide the largest reduction in overall stress field deviation. The remaining modes contribute progressively smaller corrections and may therefore be treated as secondary regulation objectives within the reduced-order framework. Consequently, the modal dynamics support the central premise of the present work: Observer and controller design may be performed in a reduced-order coordinate space while preserving the dominant behavior of the distributed electromagnetic stress field.

D. LQG Regulation Results

1) LQG Synthesis:

Using the reduced-order electromagnetic stress model developed in Section IV, a baseline Linear Quadratic Gaussian (LQG) compensator was synthesized to investigate observer-based regulation of the dominant retained Maxwell stress coordinates. The LQG architecture combines Linear Quadratic Regulator (LQR) state-feedback control with Kalman state estimation, thereby providing a nominal optimal-control framework for reduced-order electromagnetic stress regulation. The LQR weighting matrices were selected as

$$\mathbf{Q} = \text{diag}(100, 10, 5, 5, 1),$$

and

$$\mathbf{R} = 1,$$

thereby assigning the largest penalty to deviations in the dominant reduced-order Maxwell stress coordinate while imposing progressively smaller penalties on the auxiliary pinned flux, thermal, and electrostatic states. The resulting weighting hierarchy reflects the primary objective of electromagnetic stress regulation while limiting excessive control effort. Solving the continuous-time algebraic Riccati equation yielded the optimal feedback gain $\mathbf{K}_{LQR}$, while the observer gain matrix $\mathbf{L}$ was obtained from the assumed process- and measurement-noise covariance matrices. The resulting LQG compensator was subsequently used to evaluate observer-based regulation performance over a bounded set of perturbation initial conditions.

2) Closed-Loop Stability and Observer Convergence:

TABLE II
CLOSED-LOOP AND OBSERVER EIGENVALUES

Mode	$\lambda(\mathbf{A} - \mathbf{BK}_{LQR})$	$\lambda(\mathbf{A} - \mathbf{LC})$
1	-14.1323	-10.0499
2	$-0.6115 + 1.5631j$	$-1.0459 + 1.6731j$
3	$-0.6115 - 1.5631j$	$-1.0459 - 1.6731j$
4	-0.2615	-1.0112
5	-0.1500	-0.4041

The eigenvalues of both the closed-loop plant matrix $\mathbf{A} - \mathbf{BK}_{LQR}$ and observer matrix $\mathbf{A} - \mathbf{LC}$ were found to lie in the open left-half complex plane, indicating asymptotic stability of the linearized closed-loop dynamics and convergence of the state estimator. The resulting eigenvalues are summarized in Table II. The observer poles are located further to the left than the corresponding closed-loop poles, producing estimation error dynamics that decay more rapidly than the associated plant dynamics. Consequently, the observer error dynamics are asymptotically stable, implying convergence of the estimated states toward the true plant states.

3) Initial Condition Sensitivity Analysis:

TABLE III
INITIAL CONDITION PERTURBATION BOUNDS FOR THE
1000-REALIZATION LQG SENSITIVITY STUDY

State	Lower Bound	Upper Bound
q	-0.30	+0.30
$\dot{q}$	-0.15	+0.15
$\delta\Phi_p$	-0.10	+0.10
δT	-0.08	+0.08
δV_e	-0.15	+0.15

*All simulations were performed over a fixed horizon of $t_f = 10s$.

To assess sensitivity to initial condition variation, the initial condition bounds were selected to define a moderate nondimensional perturbation neighborhood about the nominal pinned flux operating point. Since the present study is simulation-driven and does not correspond to a specific fabricated geometry or material stack, the bounds are not interpreted as material failure limits or experimentally measured operating ranges. Instead, they define a consistent local region over which the reduced-order model, observer, and controller are evaluated. The selected bounds allow appreciable variation in the dominant stress coordinate and modal velocity while keeping pinned flux, thermal, and electrostatic perturbations smaller than the normalized equilibrium values, consistent with operation near the assumed linearization point.

One thousand initial conditions were sampled uniformly from the bounded perturbation region defined in Table III. The dominant reduced-order Maxwell stress coordinate was permitted to vary by up to $\pm 30\%$ of its normalized operating value, while smaller perturbations were applied to the modal velocity, pinned flux, thermal, and electrostatic states to maintain consistency with the local reduced-order operating regime.

Convergence criteria for the stress coordinate and modal velocity were selected as

$$|q(t_f)| < 10^{-2},$$

and

$$|\dot{q}(t_f)| < 10^{-2}.$$

A trajectory was considered converged if both were satisfied at $t_f = 10s$. Under these criteria, only 3.8% of the nonlinear uncontrolled trajectories converged within the 10-second simulation horizon, whereas 100% of the LQG-controlled trajectories satisfied the convergence requirement.

Because the present study employs a normalized reduced-order model and representative parameter values, these percentages should not be interpreted as universal performance metrics. Rather, they provide a numerical indication of the improved regulation consistency obtained through observer-based feedback control over the examined perturbation set.

4) Phase Plane Regulation Behavior:

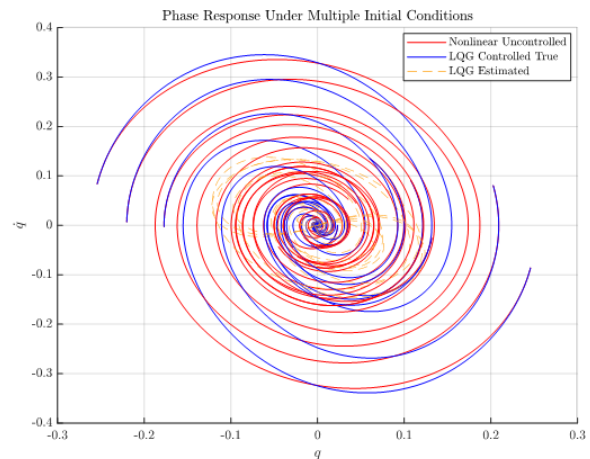


Fig. 9. Representative trajectories in the $(q, \dot{q})$ phase plane for the dominant reduced-order Maxwell stress mode obtained from the initial condition sensitivity study

Figure 9 presents phase portraits corresponding to representative trajectories drawn from the initial condition sensitivity analysis of the previous section. The nonlinear uncontrolled trajectories exhibit large phase-space excursions and slow convergence toward the equilibrium, whereas the LQG-controlled trajectories contract toward the equilibrium region and satisfy the prescribed convergence criteria across the examined perturbation set. The estimated trajectories converge toward the corresponding controlled trajectories, consistent with the stable observer dynamics established in Section VII.

5) Summary of Baseline LQG Regulation Performance:

The preceding results demonstrate that the reduced-order electromagnetic stress model admits a complete observer-based optimal-control architecture. The LQG compensator successfully regulates the dominant retained Maxwell-stress coordinate while maintaining stable observer dynamics and consistent convergence across the examined perturbation set. The phase-plane trajectories further illustrate contraction of the controlled dynamics toward the nominal pinned flux operating point and convergence of the estimated states toward the corresponding plant states.

While these results establish the feasibility of observer-based electromagnetic stress regulation, the LQG formulation is fundamentally based on a linearized representation of the underlying multiphysics dynamics. Consequently, regulation performance may degrade in the presence of parameter uncertainty, thermal drift, flux-creep effects, unmodeled dynamics, and stress field perturbations that depart from the nominal operating conditions used for controller synthesis.

To quantify the magnitude of these uncertainties and establish a basis for robust controller design, a Monte Carlo uncertainty analysis was performed. The resulting uncertainty estimates are subsequently used to determine a conservative upper bound on the lumped disturbance term required for Sliding Mode Control synthesis.

E. Monte Carlo Uncertainty Analysis

To assess robustness and obtain a conservative estimate of model uncertainty, a Monte Carlo parameter sweep is performed [21], [22]. The baseline pinned flux pressure, modal amplitudes, thermal drift coefficient, and measurement noise level are varied independently within prescribed uncertainty intervals representing plausible deviations from nominal operating conditions. For each realization, a new snapshot ensemble is generated and projected onto the nominal retained POD basis. The resulting reduced-order trajectories are compared against the nominal reduced-order dynamics to quantify the mismatch induced by parameter uncertainty and stochastic disturbances. In particular, the modal acceleration discrepancy

$$\Delta(\mathbf{x}, t) = \ddot{\mathbf{q}}_{mc}(t) - \ddot{\mathbf{q}}_{nom}(t) \quad (40)$$

is evaluated throughout the simulation horizon. The maximum observed discrepancy over all realizations is used to estimate a conservative lumped uncertainty bound

$$\Delta_{max} = \max |\Delta(\mathbf{x}, t)| \quad (41)$$

By construction, Δ_{max} represents an empirical upper bound on the retained-mode acceleration uncertainty induced by the considered parameter variations and stochastic disturbances and therefore serves as the uncertainty estimate employed in the sliding-mode control design.

The switching gain $\mathbf{K}_s$ is subsequently selected such that

$$\mathbf{K}_s > \Delta_{max},$$

consistent with standard sliding-mode robustness conditions [15]–[17], thereby ensuring that the switching term dominates bounded disturbances, parameter variation, thermal drift, and unmodeled multiphysics effects.

In addition to estimating the lumped uncertainty bound Δ_{max} , supplementary Monte Carlo sensitivity studies were performed to assess the influence of controller tuning parameters on closed-loop performance. For the LQR regulator, the diagonal state-weighting matrix $\mathbf{Q}$ was varied over a bounded range about its nominal values in order to evaluate the tradeoff between convergence rate and control effort.

For the sliding-mode controller, the sliding-surface parameter η and boundary-layer thickness φ were similarly perturbed within prescribed intervals to quantify their influence on convergence behavior, robustness, and chattering attenuation. The selected nominal values correspond to parameter regions exhibiting stable closed-loop operation and consistent regulation performance across the uncertainty ensemble, indicating that the reported results are not dependent upon a single finely tuned controller configuration.

F. Sliding Mode Control Results

Using the uncertainty estimate obtained from the Monte Carlo analysis, the Sliding Mode Control switching gain was selected to satisfy

$$\mathbf{K}_s > \Delta_{max}.$$

This selection ensures that the switching term dominates the estimated lumped uncertainty over the examined perturbation set. A boundary layer saturation function was used in place of an ideal discontinuous sign function to reduce chattering in the numerical implementation.

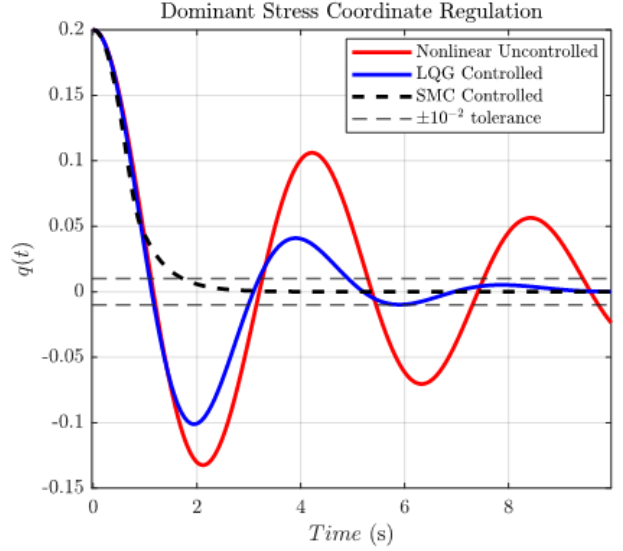


Fig. 10. Dominant reduced-order Maxwell stress coordinate $q(t)$ for the nonlinear uncontrolled, LQG-controlled, and SMC-controlled responses

Figure 10 compares the dominant reduced-order Maxwell stress coordinate under nonlinear uncontrolled, LQG-controlled, and SMC responses. The uncontrolled response exhibits slowly decaying oscillatory behavior, whereas the feedback-controlled trajectories drive the dominant stress coordinate toward the nominal pinned flux equilibrium. The LQG controller provides a nominal observer-based optimal-control baseline, while the SMC controller incorporates the Monte Carlo uncertainty estimate directly into the robust control law.

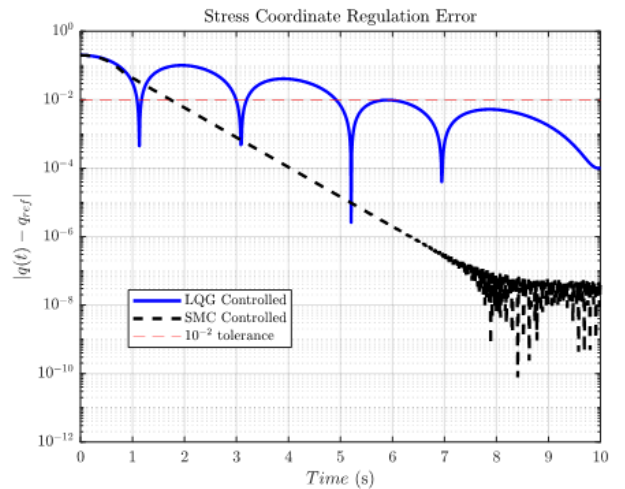


Fig. 11. Absolute stress coordinate regulation error for the LQG and SMC controllers

Figure 11 shows the corresponding absolute regulation error for the LQG and SMC responses. The dashed horizontal red

line denotes the prescribed terminal convergence tolerance. The results demonstrate that the SMC formulation achieves bounded stress coordinate regulation under the conservative uncertainty estimate obtained from the Monte Carlo sweep. Together, these results support the use of SMC as a robust extension of the baseline LQG architecture for reduced-order electromagnetic stress regulation under uncertainty.

G. Discussion of Results

The simulation results demonstrate that the proposed electromagnetic stress regulation framework admits a compact reduced-order representation while preserving the dominant dynamic characteristics of the distributed stress field. The POD analysis showed that a small number of retained modes captured the majority of the electromagnetic stress energy, supporting the use of reduced-order modeling for observer and controller synthesis.

The resulting reduced-order model was shown to be stabilizable and observable, enabling construction of an observer-based Linear Quadratic Gaussian (LQG) compensator. Closed-loop simulations demonstrated successful regulation of the dominant electromagnetic stress coordinate and convergence of the corresponding state estimates. The phase-plane trajectories further illustrate contraction of the controlled dynamics toward the nominal operating point and consistency of the observer dynamics.

Monte Carlo analysis indicated that the retained-mode uncertainty remained bounded over the bounded perturbation neighborhood and provided a systematic means of estimating a conservative lumped uncertainty bound. This uncertainty estimate was subsequently incorporated into the Sliding Mode Control design through selection of the switching gain.

Comparative simulations demonstrated that both the LQG and SMC architectures achieved regulation of the dominant electromagnetic stress coordinate. However, the SMC formulation explicitly incorporated the uncertainty estimates obtained from the Monte Carlo study and therefore provides a more robust control architecture in the presence of parameter variation, thermal drift, stochastic disturbances, and unmodeled dynamics. Taken together, these results indicate that electromagnetic stress regulation can be formulated within a unified reduced-order control framework that combines modal decomposition, state estimation, uncertainty quantification, and robust feedback control.

While the present study is based on a normalized simulation model, the methodology establishes a systematic pathway from distributed electromagnetic field descriptions to tractable observer-based and robust control architectures suitable for future high-energy superconducting and hybrid electromagnetic systems.

VIII. FUTURE RESEARCH DIRECTIONS

A. Advanced Reduced-Order Basis Functions

The present work employs POD-derived empirical basis functions obtained directly from electromagnetic stress snapshots. A natural future extension is the incorporation of physically motivated orthogonal bases, including Zernike and polar Jacobi polynomial expansions, which may provide additional geometric interpretability and facilitate direct incorporation of known symmetry properties of circular electromagnetic stress domains. Hybrid spectral-POD approaches may further improve model compactness while preserving meaningful physical structure.

B. Stress/Strain-Assisted Quantum Dot Tuning

The electromagnetic stress-regulation framework developed herein may be extended to electromechanical strain fields in semiconductor nanostructures. In such settings, the controlled state variables would no longer represent dominant electromagnetic stress modes, but rather reduced-order strain and confinement modes governing the electronic properties of the device. Because strain directly modifies the local crystal potential landscape, controlled deformation may be used to tune quantum confinement energy, bandgap structure, valley splitting, excitonic transitions, and optical emission characteristics. A natural extension of the present framework is the incorporation of electromechanical coupling, allowing simultaneous regulation of electromagnetic and strain fields through a unified observer-based control architecture. Such an approach could enable dynamic tuning of carrier confinement, band structure, valley splitting, and optical transition energies while compensating for fabrication variability, thermal drift, and charge-noise-induced perturbations. More broadly, reduced-order observer-controller architectures may provide a systematic means by which to stabilize strain-sensitive quantum operating points in future semiconductor and hybrid quantum-device platforms.

C. Reconfigurable Superconducting Networks and Field Topologies

The reduced-order methodology may also be extended to distributed superconducting networks containing multiple coupled flux states and actively reconfigurable electromagnetic field topologies. Future implementations may employ engineered superconducting, magnetic, or hybrid materials whose effective field distributions can be modified through distributed actuation and feedback regulation. Within such systems, reduced-order state variables could describe collective flux configurations, network-level electromagnetic modes, or distributed electromagnetic field configurations. Observer-based feedback control may then be employed to regulate collective network dynamics, compensate for flux creep and parameter drift, and enable adaptive reconfiguration of operating conditions in response to environmental disturbances or changing performance requirements. Such architectures represent a natural extension of the present single-domain formulation toward distributed electromagnetic control systems.

D. Adaptive Confinement Landscapes

The present framework may be extended to dynamically reconfigurable confinement potentials in semiconductor nanostructures. In this setting, reduced-order state variables could describe dominant confinement-energy modes, while electrostatic gate arrays actively reshape the local potential landscape to maintain desired operating conditions in the presence of charge noise, fabrication imperfections, and environmental drift. By combining reduced-order state estimation with adaptive feedback control, future architectures may be capable of continuously regulating confinement profiles, tunnel-coupling parameters, and quantum operating points in real time. Such capabilities may prove valuable for scalable semiconductor quantum information platforms, where robust maintenance of quantum operating conditions remains a significant challenge.

E. Extension to Multi-Modal Observer-Based Control

The present investigation intentionally restricts the feedback controller to regulation of the dominant retained POD coordinate in order to establish the feasibility of the proposed reduced-order electromagnetic stress regulation framework. Although this single-mode formulation captures the majority of the electromagnetic stress energy in the representative snapshot ensemble, the methodology naturally extends to simultaneous regulation of the complete retained modal coordinate vector.

In the multi-modal formulation, the reduced-order state vector would be augmented to include the additional retained modal coordinates and their corresponding velocities,

$$\mathbf{x}(t) = \begin{bmatrix} q_1(t) \\ \dot{q}_1(t) \\ q_2(t) \\ \dot{q}_2(t) \\ q_3(t) \\ \dot{q}_3(t) \\ \Phi_p(t) \\ T(t) \\ \mathbf{V}_{seg}(t) \end{bmatrix} \quad (42)$$

where $\mathbf{V}_{seg}$ denotes the vector of independently addressable segmented-electrode voltages as in Section IV-A. Consequently, the nonlinear state equations, observer dynamics, and corresponding linearized state-space model would increase in dimension, producing higher-order system matrices together with multi-input, multi-output (MIMO) control and estimation architectures.

Rather than a single equivalent electrostatic control input, future implementations may employ independently controlled segmented electrodes whose voltages are synthesized through coordinated modal feedback. The associated control law would therefore determine the spatial distribution of electrode actuation required to regulate multiple retained electromagnetic stress modes simultaneously, providing direct control over increasingly complex stress field topologies. In this setting,

the equivalent scalar input of the present work is replaced by a vector of independently addressable commanded electrode voltages, namely

$$\mathbf{u}(t) = [u_1 \ u_2 \ \dots \ u_n]^T,$$

and the state-space matrices are also expanded to accommodate the additional retained modal coordinates.

Future investigations may additionally incorporate robust sliding-mode observers in place of the baseline Kalman estimator to improve state reconstruction under stronger modeling uncertainty, parameter variation, and measurement disturbances [18], [20]. As the reduced-order model is extended to multiple retained modes and independently addressable segmented electrodes, the resulting formulation naturally evolves into a higher-dimensional MIMO control problem. Sliding-mode methodologies have been successfully extended to increasingly complex nonlinear and hybrid dynamical systems [16], [17], [19], suggesting that analogous robust observer-controller architectures may provide an effective framework for multi-modal electromagnetic stress regulation while preserving the computational efficiency afforded by Proper Orthogonal Decomposition.

IX. POTENTIAL EXTENSIONS TO QUANTUM SEMICONDUCTOR SYSTEMS

Although the present work focuses on electromagnetic stress regulation in pinned flux superconducting systems, the underlying reduced-order modeling and observer-based control methodology is not restricted to macroscopic electromagnetic states. Despite the underlying quantum-mechanical complexity, many semiconductor quantum devices are effectively governed by a low-dimensional set of experimentally tunable parameters, including confinement energy, detuning, tunnel coupling, spin splitting, valley splitting, and proximity-induced superconducting gap. Consequently, the reduced-order framework developed herein may provide a foundation for observer-based estimation and feedback control of quantum operating states in future hybrid superconducting-semiconductor architectures.

A. Reduced-Order State Representations for Quantum Semiconductor Devices

The same reduced-order control philosophy can be applied to quantum semiconductor devices by replacing macroscopic stress modes with reduced-order quantum device modes, such as confinement energy, detuning, tunnel coupling, Zeeman splitting, valley splitting, and proximity-induced superconducting gap. In this setting, flux pinning provides a stable magnetic or superconducting boundary condition, while electrostatic gates provide high-bandwidth control authority. Similar to the dominant electromagnetic stress coordinates retained through Proper Orthogonal Decomposition (POD) in the present work, many quantum semiconductor devices admit low-dimensional

descriptions capturing the dominant behavior of a substantially larger underlying physical system. Reduced-order coordinates derived from experimentally observable quantities may therefore provide a tractable state representation for observer and controller synthesis while avoiding the computational complexity associated with full quantum-mechanical device models.

B. Candidate Hybrid Superconducting-Semiconductor Architectures

1) *Pinned Flux-Stabilized Quantum Dot Arrays*: Pinned magnetic flux structures may provide stable local magnetic boundary conditions for coupled quantum-dot arrays, while electrostatic gate actuation supplies high-bandwidth tuning of confinement and detuning states. In such architectures, the persistent magnetic environment established by flux pinning could reduce sensitivity to external magnetic-field fluctuations while preserving localized control authority through gate-defined potentials. Reduced-order state variables describing confinement energy, inter-dot detuning, tunnel coupling, and spin-splitting parameters could then be estimated and regulated using observer-based feedback similar to the observer-based architectures developed herein. By combining passive magnetic stabilization with active electrostatic control, such systems may enable improved operating-point stability, enhanced robustness to charge-noise perturbations, and adaptive regulation of coupled quantum-dot states within scalable semiconductor quantum-device platforms.

2) *Hybrid Superconducting-Semiconductor Dot Controller*: Future hybrid superconducting-semiconductor devices may benefit from observer-based robust control architectures capable of regulating quantum operating states in the presence of charge noise, flux drift, thermal variation, and fabrication-induced parameter uncertainty. In such systems, reduced-order state variables could include quantum-dot detuning, tunnel coupling, spin-splitting energies, valley splitting, and proximity-induced superconducting gap parameters. Because many of these quantities are only indirectly observable through available electrical measurements, state-estimation techniques such as Extended Kalman Filtering (EKF), Unscented Kalman Filtering (UKF), or nonlinear observer formulations may be employed to reconstruct the underlying device state in real time. These estimates could then be incorporated into robust feedback-control laws, including Sliding Mode Control architectures, to regulate quantum operating points and maintain desired device characteristics despite environmental disturbances and model uncertainty. Such an approach may provide a systematic framework for adaptive stabilization and control of next-generation hybrid quantum-semiconductor platforms.

C. Observer-Based Quantum Operation Point Stabilization

Many quantum-semiconductor devices operate most effectively within narrow regions of parameter space commonly referred to as sweet spots, where sensitivity to charge noise,

magnetic-field fluctuations, fabrication variability, and environmental disturbances is minimized. Examples include optimal bias points in superconducting qubits, charge configurations in semiconductor quantum dots, and operating regimes in hybrid superconducting-semiconductor devices where coherence times and device stability are enhanced.

In practice, however, many of the physical quantities governing device behavior are not directly measurable in real time. Tunnel coupling strengths, valley splitting energies, localized charge distributions, induced superconducting gap parameters, and other internal device variables often influence system performance while remaining only indirectly observable through available electrical measurements. Consequently, maintaining operation near a desired quantum operating point becomes both an estimation and control problem.

The observer-based framework developed in the present work provides one possible approach for addressing this challenge. Reduced-order observers may be employed to reconstruct hidden device states from available measurements, thereby providing real-time estimates of quantities that cannot be directly sensed. These estimates may then be incorporated into feedback-control laws designed to regulate confinement potentials, electrostatic gate voltages, magnetic boundary conditions, or other control inputs influencing the quantum operating state.

Within such a framework, observer-based feedback could potentially compensate for slow thermal drift, charge-noise fluctuations, device aging effects, and fabrication-induced parameter variations. Although the present investigation focuses on electromagnetic stress regulation in pinned flux superconducting systems, the underlying estimation-and-control methodology may provide a useful foundation for future stabilization of quantum operating points in hybrid superconducting-semiconductor architectures.

X. CONCLUSION AND OUTLOOK

Although the concepts discussed in the previous section remain exploratory, they illustrate how the reduced-order estimation-and-control methodology developed in this work may extend beyond electromagnetic stress regulation and into a broader class of superconducting, semiconductor, and hybrid quantum-device systems. Continued development of reduced-order state representations, observer-based estimation techniques, uncertainty quantification methods, and robust feedback-control architectures may provide a pathway toward adaptive regulation of increasingly complex electromagnetic and quantum-mechanical operating states.

REFERENCES

- [1] G. Schweitzer and E. H. Maslen, *Magnetic Bearings: Theory, Design, and Application to Rotating Machinery*. Berlin, Germany: Springer, 2009.
- [2] H.-S. Han, *Magnetic Levitation: Maglev Technology and Applications*. Dordrecht, Netherlands: Springer, 2014.
- [3] A. A. Abrikosov, "On the Magnetic Properties of Superconductors of the Second Group," *Sov. Phys. JETP*, vol. 5, no. 6, pp. 1174–1182, Dec. 1957.
- [4] J. Bardeen and M. J. Stephen, "Theory of the Motion of Vortices in Superconductors," *Phys. Rev.*, vol. 140, no. 4A, pp. A1197–A1207, Nov. 1965.
- [5] E. H. Brandt, "The Flux-Line Lattice in Superconductors," *Rep. Prog. Phys.*, vol. 58, no. 11, pp. 1465–1594, Nov. 1995.
- [6] M. Tinkham, *Introduction to Superconductivity*, 2nd ed. New York, NY, USA: McGraw-Hill, 1996.
- [7] M. Rocci, G. De Simoni, M. Chirilli, F. Paolucci, E. Strambini, F. Giazotto, and C. Puglia, "Large Enhancement of Critical Current in Superconducting Films Produced by Electrostatic Fields," *Adv. Quantum Technol.*, vol. 4, no. 12, Art. no. 2100624, Dec. 2021.
- [8] F. Deschaux, F. Gouaisbaut, and Y. Ariba, "Nonlinear Control for an Uncertain Electromagnetic Actuator," *arXiv:1810.00770*, 2018.
- [9] L. Sirovich, "Turbulence and the Dynamics of Coherent Structures. Part I: Coherent Structures," *Quarterly of Applied Mathematics*, vol. 45, no. 3, pp. 561–571, 1987.
- [10] P. Holmes, J. L. Lumley, G. Berkooz, and C. W. Rowley, *Turbulence, Coherent Structures, Dynamical Systems and Symmetry*, 2nd ed. Cambridge, U.K.: Cambridge Univ. Press, 2012.
- [11] Y. C. Liang, H. P. Lee, S. P. Lim, W. Z. Lin, K. H. Lee, and C. G. Wu, "Proper Orthogonal Decomposition and Its Applications—Part I: Theory," *J. Sound Vib.*, vol. 252, no. 3, pp. 527–544, May 2002.
- [12] A. C. Antoulas, *Approximation of Large-Scale Dynamical Systems*. Philadelphia, PA, USA: SIAM, 2005.
- [13] R. E. Kalman, "A New Approach to Linear Filtering and Prediction Problems," *J. Basic Eng.*, vol. 82, no. 1, pp. 35–45, Mar. 1960.
- [14] D. G. Luenberger, "Observers for Multivariable Systems," *IEEE Trans. Autom. Control*, vol. 11, no. 2, pp. 190–197, Apr. 1966.
- [15] J.-J. E. Slotine and W. Li, *Applied Nonlinear Control*. Englewood Cliffs, NJ, USA: Prentice Hall, 1991.
- [16] V. I. Utkin, *Sliding Modes in Control and Optimization*. Berlin, Germany: Springer-Verlag, 1992.
- [17] C. Edwards and S. K. Spurgeon, *Sliding Mode Control: Theory and Applications*. London, U.K.: Taylor Francis, 1998.
- [18] S. Drakunov and V. Utkin, "Sliding Mode Observers: Tutorial," in *Proc. 34th IEEE Conf. Decis. Control*, New Orleans, LA, USA, 1995, vol. 4, pp. 3376–3378.
- [19] S. Drakunov, M. Dogruel, and U. Ozguner, "Sliding Mode Control in Hybrid Systems," in *Proc. IEEE Int. Symp. Intell. Control*, Chicago, IL, USA, 1993, pp. 186–189.
- [20] B. L. Walcott and S. H. Zak, "State Observation of Nonlinear Uncertain Dynamical Systems," *IEEE Trans. Autom. Control*, vol. 32, no. 2, pp. 166–170, Feb. 1987.
- [21] R. Y. Rubinstein and D. P. Kroese, *Simulation and the Monte Carlo Method*, 2nd ed. Hoboken, NJ, USA: Wiley, 2008.
- [22] J. C. Helton and F. J. Davis, "Latin Hypercube Sampling and the Propagation of Uncertainty in Analyses of Complex Systems," *Reliability Engineering System Safety*, vol. 81, no. 1, pp. 23–69, Jul. 2003.
- [23] K. Willcox and J. Peraire, "Balanced Model Reduction via the Proper Orthogonal Decomposition," *AIAA J.*, vol. 40, no. 11, pp. 2323–2330, Nov. 2002.
- [24] G. Blatter, M. V. Feigel'man, V. B. Geshkenbein, A. I. Larkin, and V. M. Vinokur, "Vortices in High-Temperature Superconductors," *Rev. Mod. Phys.*, vol. 66, no. 4, pp. 1125–1388, Oct. 1994.
- [25] L. Burlachkov, "Magnetic Relaxation over the Bean-Livingston Surface Barrier," *Phys. Rev. B*, vol. 47, no. 13, pp. 8056–8060, Apr. 1993.